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About the UltraWave24 MWSource : Features

Features

The MWSource generates high-accuracy, low-noise, and low-distortion sine waves. It performs automatic leveling (amplitude level calibration) at each programmed frequency. In addition, the

MWSource controls the DUT Clock.

CW Source

Basic functional specifications are:

- Frequency range: 50MHz to 6GHz
- Amplitude range:
  - 50MHz to 2.7GHz: -130dBm to 13dBm
  - 2.7GHz to 6GHz: -130dBm to 10dBm
- Amplitude resolution: 0.01dB

DUT Clock

Basic functional specifications are:

- 8:1 splitter with independent output relays
- Frequency range: 8MHz to 105MHz
- Frequency resolution: 100Hz
- Amplitude range: -20 dBm to 13 dBm
- Amplitude resolution: <0.25dB

For UltraWave24 hardware specifications, see UltraWave24 Specifications.

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About the UltraWave24 MWSource : Safety Information

Safety Information

When operating the Microwave Source instrument, always follow general UltraFLEX test system safety guidelines. See Test System Safety.

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About the UltraWave24 MWSource : Key Terms

Key Terms

See Key Terms in About the UltraWave24 Subsystem.

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UW24SrcDefTopic

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About the UltraWave24 MWSource

About the UltraWave24 MWSource

The MWSource is a programmable CW source that generates signals in the frequency range from 50MHz to 6GHz.

The MWSource logical instrument is also the programming interface for the DUT Clock on a Synthesizer Reference board.

Applications for the MWSource include prescalers, RF amplifiers, TV tuners, cellular and connectivity radios, IF devices, and operational amplifiers.

In addition to the MWSource, the UltraWave24 subsystem supports several other types of sources:

MWModulatedSource	Source modulated by an arbitrary waveform. See About the UltraWave24 MWModulatedSource.
MWMultitoneSource	Multitone source for two or more tones. See About the UltraWave24 MWMultiToneSource.
MWNoiseSource	Noise source. See About the UltraWave24 MWNoiseSource.

This section includes the following topics:

- Features
- Safety Information
- Key Terms

MW Source information includes the following sections:

- UltraWave24 MWSource Theory of Operation
- UltraWave24 MWSource Programming
- UltraWave24 MWSource Debug Display

Related Topics

- UltraWave24 Specifications
- MWSource (VBT syntax)
- UltraWave24 DIB Design Guidelines
- About the UltraWave24 Subsystem
- UltraWave24 Examples

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UltraWave24 MWSource Debug Display

UltraWave24 MWSource Debug Display

The UltraWave24 Microwave Debug Display can display one or more instrument blocks for the MWSource logical instrument, each showing the state of a MWSource in the context of a site.

This section includes the following topic:

- UltraWave24 MWSource Debug Display Controls

For an overview of all displays supporting the UltraWave24 subsystem, see UltraWave24 Displays.

For more information on using debug displays, see the following:

- Using Debug Displays in TDE
- Producing Code from Debug Displays

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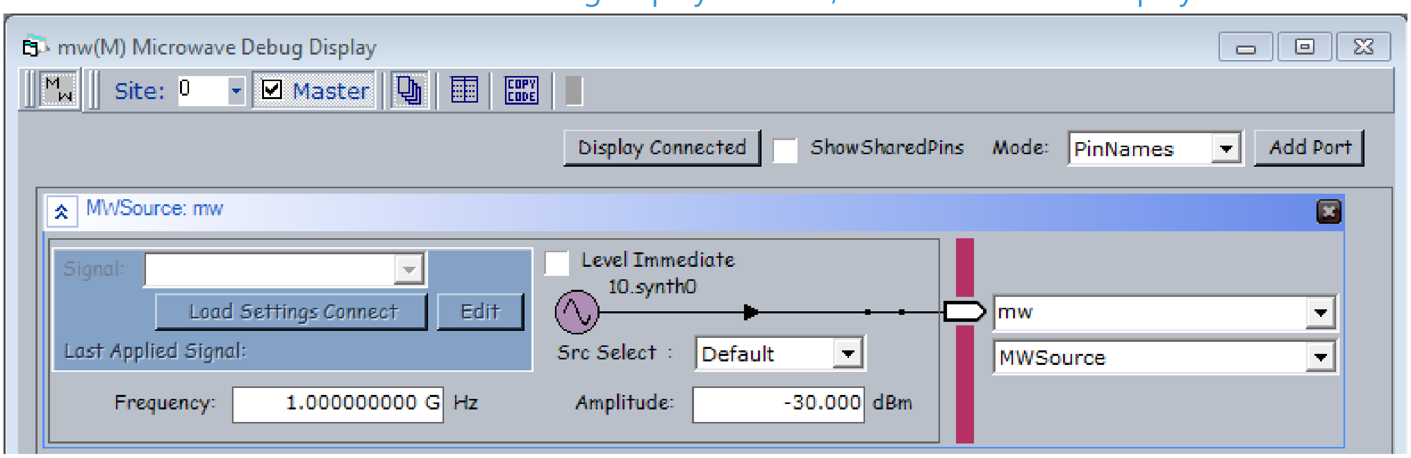
UltraWave24 MWSource Debug Display : UltraWave24 MWSource Debug Display Controls

UltraWave24 MWSource Debug Display Controls

The display controls for the MWSource assigned to a port allow you to view and set instrument parameters, which are also accessible through the Pins interface of the MWSource VBT language. The display shows DUT Clock pins on the MW Synthesizer HD board.

For information about the common debug display controls, see UltraWave24 Displays.



MWSource Logical Instrument Display

The following controls are available on the MWSource logical instrument block:

Frequency	Displays and sets the source frequency. The tooltip displays the allowable frequency range.
Src Select	Displays and sets the synthesizer used to generate source signals, Primary, Secondary, or Default.
Amplitude	Displays and sets the source amplitude. The tooltip displays the allowable amplitude range.

Level Immediate	Sets the MWSource leveling mode to Level Immediate. See MWSource.Pins.LevelImmediate in the VBT Syntax Reference.
synth label	Identifies the synthesizer (slot and number) associated with this Microwave Source logical instrument.
Signal	Name of the signal used by the source. Available Signals appear on the drop-down list. If a default signal has been defined, the default signal's name appears automatically in this field. If the menu contains no signal names, no signals were previously defined in VBT. The MWSource can operate without a signal.
Edit	Brings up the PSet and Signals Editor (PSSE) with the selected signal loaded. This control is grayed out if no signals are available. See PSet and Signals Editor in the TDE section.
Load Settings Connect	Loads the selected signal's timing settings as specified on the Mixed Signal Timing sheet, as well as all of the instrument's settings and connects the MWSource.
Info	Displays information for the selected Signal in a pop-up balloon window.
Last Applied Signal	Name of the last signal applied to the MWSource.
Connection relay	The relay control connects or disconnects the MWSource logical instrument to the specified pin or channel. Click the relay to open or close the connection.

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UltraWave24 MWSource Programming

UltraWave24 MWSource Programming

The MWSource is a MW logical instrument in the IG-XL environment. In general, you program the MWSource like the other MW instruments.

The following topics are available:

- Sourcing Sine Waves Using the MWSource
- MWSource Pattern Control

Before writing a program using the MWSource, see UltraWave24 Instrument Programming.

For more information, see:

- MWSource (VBT Syntax Reference)
- UltraWave24 Examples

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UltraWave24 MWSource Programming : Sourcing Sine Waves Using the MWSource

Sourcing Sine Waves Using the MWSource

This section includes the following topics:

- Programming Concepts
- Programming Steps
- Example

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UltraWave24 MWSource Programming : Sourcing Sine Waves Using the MWSource : Programming Concepts

Programming Concepts

Source Leveling

Source level calibration, also called leveling, finely calibrates the MWSource at each different frequency set in a program. Leveling takes place at run time.

Leveling occurs the first time a specific frequency is set for the MW Source. Later VBT statements setting the same frequency use cached data and do not level, unless leveling is invoked using the Pins.LevelImmediate VBT property for the source type. For example, see the LevelImmediate property for the Pins interface of TheHdw.MWSource.

When you call a MWSource from a pattern, level calibration occurs for all parameter value set combinations before the pattern starts.

MWSource Reset Values

The table lists the local reset values for the MWSource.

MWSource Reset Values

Setting	Reset Value
Amplitude	-30dBm
Frequency	1.0GHz
Connections	Disconnected

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UltraWave24 MWSource Programming : Sourcing Sine Waves Using the MWSource : Programming Steps

Programming Steps

The MWSource is a simple CW source instrument producing a sine wave at a programmed frequency and amplitude.

To source a sine wave to an assigned RF port:

1.	Set source signal amplitude.

2.	Set source signal frequency.

3.	Connect to the DUT pin or channel.

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UltraWave24 MWSource Programming : Sourcing Sine Waves Using the MWSource : Example

Example

```
Function MWSourceSignal(ByVal RF_Frequency As Double, _
                ByVal input_power As Double, _
                ByVal Source_Pin As String)

' Set up Source
TheHdw.MWSource.Pins(Source_Pin).Frequency = RF_Frequency
TheHdw.MWSource.Pins(Source_Pin).Amplitude = input_power

' Connect MWSource
TheHdw.MWSource.Pins(Source_Pin).Connect

' Disconnect MWSource
TheHdw.MWSource.Pins(Source_Pin).Disconnect

End Function
```

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UltraWave24 MWSource Programming : MWSource Pattern Control

MWSource Pattern Control

You can control an MWSource instrument using pattern microcodes, allowing changes to instrument states and parameters during a program. You can obtain repeatable timing by programming an instrument using pattern microcodes and Signals programming.

You create a pattern independently of the test program using the PatternTool or a text editor.

Patterns are composed of a series of numbered digital vectors. These digital vectors define the data for each channel listed in the pattern, as well the pattern microcode used for controlling the instrument during pattern execution.

The following topics are available:

- MWSource Pattern Microcodes
- MWSource ASCII Pattern Syntax

☑ **Note:**

To use pattern microcodes to control an MWSource on UltraFLEX, you must include a digital pin in the pin map, the channel map, and the pin list of the pattern module.

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UltraWave24 MWSource Programming : MWSource Pattern Control : MWSource Pattern Microcodes

MWSource Pattern Microcodes

A microcode controls a single function, such a starting or stopping the sourcing of a signal. Extended microcode commands are available for use inside a pattern. Patterns allow you to control the precise timing of when a signal is sourced. To establish repeatable timing, or to synchronize with other instruments (such as the digital subsystem) in the tester, use a pattern. If you want to start generating a signal but are not concerned with precise timing, however, then you do not need a pattern.

The table lists the extended microcodes used to issue commands to an MWSource instrument from a pattern.

MWSource Pattern Microcodes

Microcode	Functional Description
LoadSettingsConnect <SignalName>	Instructs an MWSource to load the settings for the named capture signal and connects the hardware to the DUT. SignalName is required.
Disconnect	Instructs an MWSource to reset all connections to their disconnected state.

Note the following when using microcodes with an MWSource:

- IG-XL has rules regarding the necessary spacing in vectors between microcodes on a specific channel. If you place microcodes too close together, IG-XL will return an error. General guidelines for spacing microcodes include the following:

- LoadSettingsConnect: 400s (typical)
- Disconnect: <200s (typical)

- An MWSource does not use PSets. It provides the equivalent capability using Signals programming in VBT.
- When using an UltraWave logical instrument (such as an MWSource), any vector that contains a POP microcode must be fully defined across all defined logical instrument instances.
  - IG-XL returns an error if you do not follow this guideline.
  - Reapplying the same Signal to a given logical instrument instance results in the hardware NOPing the application. (This prevents glitches.)

For more information and an example, see Using UltraWave24 Instruments with POP.

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UltraWave24 MWSource Programming : MWSource Pattern Control : MWSource ASCII Pattern Syntax

MWSource ASCII Pattern Syntax

MWSource pattern microcodes are instructions you can place on an individual vector of a pattern. For that vector, a microcode instructs the MWSource to perform a particular action, such as starting a new signal.

You can only use MWSource microcodes with pins that the instruments statement has specified as MWSource. For example:

```
instruments = {
    DUT_AIN:MWSource;
}
```

For more information, see Instruments Statement in the Pattern Language section.

If a vector contains any MWSource related microcode other than nop, that vector is placed in SRM.

The format for specifying an MWSource microcode on a particular vector is:

(pin-list : MWSource = microcode)

The following code shows an example of ASCII pattern microcode syntax for an MWSource. The figure shows this pattern in PatternTool.

```
instruments = {
    Source:MWSource 1;
}

vm_vector
vm_MWSourceOnly_A ( $tset       , Dig1)
{
start_label MWSourceOnly:
            > BasicSource  0    ;
```

```
(Source:MWSource = LoadSettingsConnect GSM)
            > BasicSource  0    ;
repeat 10000     > BasicSource  0    ; //1ms wait time
(Source:MWSource = LoadSettingsConnect WCDMA)
            > BasicSource  0    ;
repeat 10000     > BasicSource  0    ; //1ms wait time
(Source:MWSource = Disconnect)
            > BasicSource  0    ;
repeat 2000      > BasicSource  0    ; //200us wait time
            > BasicSource  0    ;
            > BasicSource  0    ;
            > BasicSource  0    ;
            > BasicSource  0    ;
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            > BasicSource  0    ;
            > BasicSource  0    ;
    }
```

MWSourceOnly.PAT (vm_MWSourceOnly_A - VM)

TSet	Command	Label	Vector	Source (MWSource)	Digi 1	Comment
BasicSource		start_label MWSourceOnly:	0		0	
BasicSource			1	LoadSettingsConnect GSM	0	
BasicSource	repeat 10000		2		0	1mS wait time
BasicSource			3	LoadSettingsConnect WCDMA	0	
BasicSource	repeat 10000		4		0	1mS wait time
BasicSource			5	Disconnect	0	
BasicSource	repeat 2000		6		0	200uS wait time
BasicSource			7		0	
BasicSource			8		0	
BasicSource			9		0	
BasicSource			10		0	
BasicSource			11		0	
BasicSource			12		0	
BasicSource			13		0	
BasicSource			14		0	
BasicSource			15		0	

Vertical: 65 Vectors | Vectors 0 to 15

MWSource Pattern Microcodes Example

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UltraWave24 MWSource Theory of Operation

UltraWave24 MWSource Theory of Operation

The MWSource instrument produces a programmable CW signal using the sine wave output of an RF frequency synthesizer.

In addition, the MWSource logical instrument controls the DUT Clock outputs on a Synthesizer HD board.

The following topics are available:

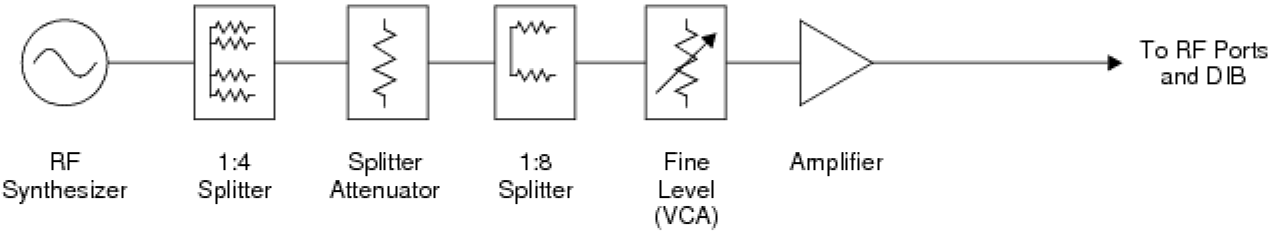
For information on MWSource instrument performance specifications and standards, see UltraWave24 Specifications.

CW Source Basic Operation

The RF synthesizer used by a MWSource generates a sine wave. The sine wave is routed through a 1:4 splitter to the attenuator and 1:8 splitter feeding the source paths on a front end module on a Measure HD board.

The frequency synthesizer is locked to a master clock reference to maintain a coherent timing relationship with the test system.

An attenuator and a front end module attenuate and amplify the signal as needed to set the programmed power level at the DIB. See the figure.



UltraWave24 MWSource Block Diagram

The front end modules use attenuators and VCAs to provide fine level adjustment. The signal path depends on the programmed amplitude. The noise level may change when an attenuator adjustment occurs due to an amplitude change.

☑ **Note:**

The signal path of a MWSource has more than one separate attenuator and amplifier. All level control elements are combined in the blocks in the diagram.

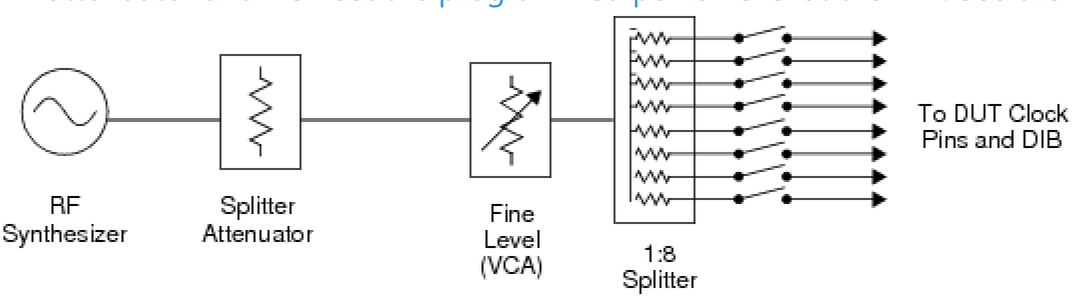
The splitters in the signal path are part of the fanout architecture of the UltraWave24. They allow multiple MWSources to connect to multiple RF ports and device pins, without splitters on the DIB.

DUT Clock Operation

The RF synthesizer used by a MWSource controlling a DUT Clock output generates a sine wave. The sine wave is routed to DUT Clock section on the Synthesizer HD board. Each section has level control and eight independent outputs with disconnect relays.

The frequency synthesizer is locked to a master clock reference to maintain a coherent timing relationship with the test system.

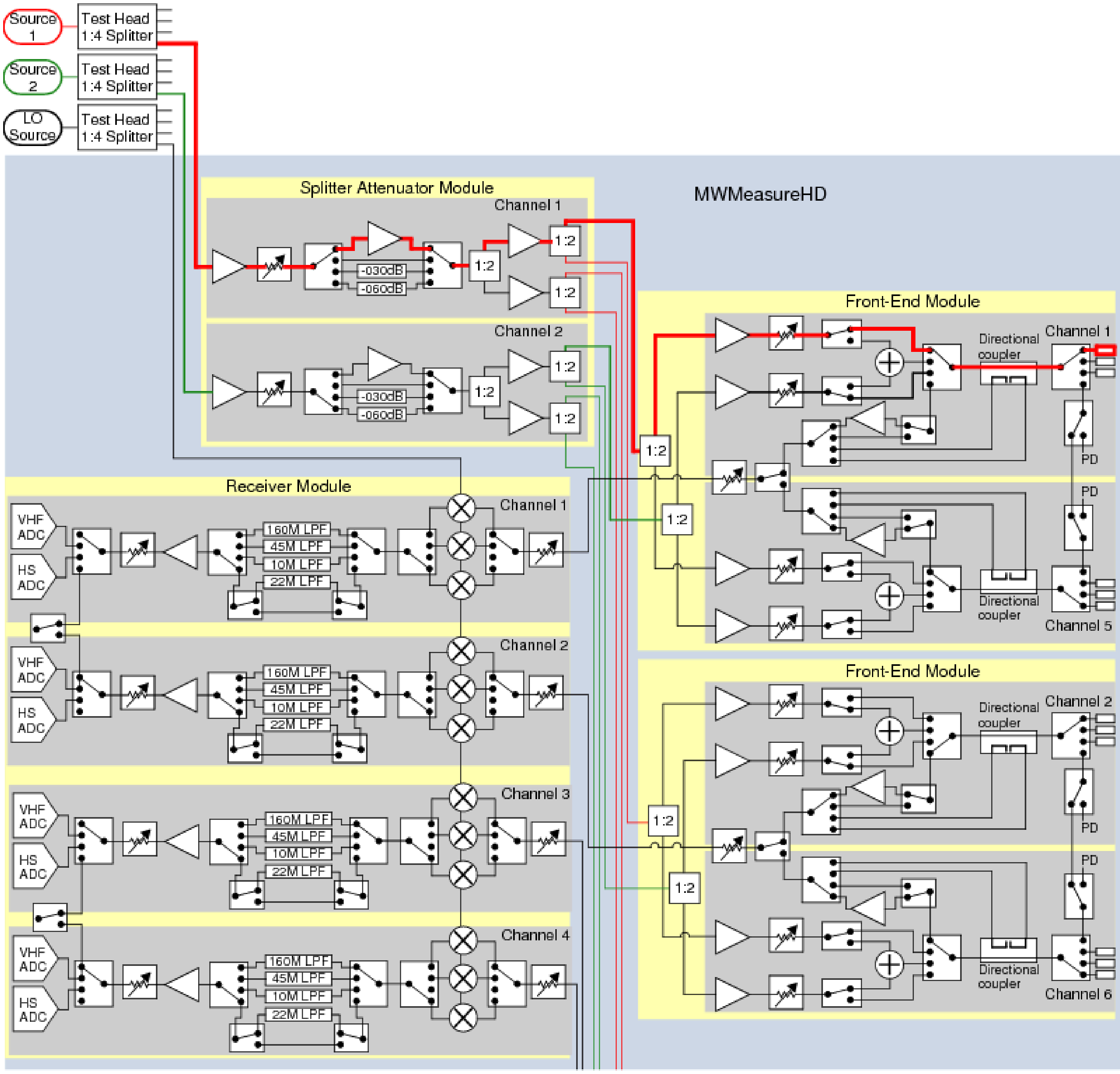
An attenuator and VCA set the programmed power level at the DIB. See the figure.



UltraWave24 DUT Clock Block Diagram

MWSource Hardware Functions and Signal Flows

The figure shows the source signal flows for the connection and operation of an MWSource providing CW signals. Heavy lines represent source signal flows. See UltraWave24 Hardware Functions for more information about the functions in the diagram.



MWSource Hardware Functions and Signal Path, Two of Four FEMs Shown

For DIB connections (MW ports, Pogo pins) and signal names for the MW subsystem and instruments, see User View of UltraWave24 SMPM Connector Blocks, User View of UltraWave24 DIB Pad Patterns, and UltraWave24 Signal Descriptions.

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